6.1 — Indefinite Integral Rules

Worked solutions. Use after the 30-minute practice set. If your answer differs, locate the first incorrect step. Equivalent forms of an antiderivative can differ by a constant; differentiate to check.

1. (a) Integrate term by term:

$$\int (8x^3 - 6x + 5)\,dx = \boxed{2x^4 - 3x^2 + 5x + C}.$$

(b) Rewrite as $3x^{1/2} - 2x^{-2}$, then raise each exponent by one and divide:

$$\boxed{2x^{3/2} + \frac{2}{x} + C}.$$

The positive $2/x$ differentiates to $-2/x^2$.

2. (a) The exponent -1 uses the logarithm rule: $\boxed{4\ln|x| + 2e^x + C}$ on any interval avoiding zero.

(b) Integrate the three terms: $\boxed{3\tan x + 2\cos x + \arctan x + C}$. Since $(\cos x)' = -\sin x$, the cosine coefficient is positive.

3. (a) The inner derivative is 3: $6 \cdot \frac{1}{3}\sin(3x) = \boxed{2\sin(3x) + C}$.

(b) The inner derivative is -2: $\boxed{-\frac{5}{2}e^{-2x} + C}$. Differentiation cancels the factor -2 in the denominator.

4. First $F = x^3 - 2x^2 + 2x + C$. From $F(1) = 5$, $1 - 2 + 2 + C = 5$, so $C = 4$ and

$$\boxed{F(x) = x^3 - 2x^2 + 2x + 4}.$$

Check: $F' = 3x^2 - 4x + 2$ and $F(1) = 1 - 2 + 2 + 4 = 5$.

5. The derivative of the proposed expression is

$$\frac{d}{dx}\left(\frac{\ln|x|}{x} + \sin x\right) = \frac{1 - \ln|x|}{x^2} + \cos x,$$

not $1/x + \cos x$. The logarithmic term should not be divided by x; the general answer also needs a constant:

$$\boxed{\ln|x| + \sin x + C}.$$

6. Integrate velocity: $s(t) = 3t^2 - 4t + C$. Since $s(0) = 7$, $\boxed{s(t) = 3t^2 - 4t + 7}$ meters.

$$\boxed{s(3) = 22 \text{ m}}, \qquad \boxed{s(3) - s(0) = 15 \text{ m}}.$$

Position includes the initial location; displacement measures only the change from that location.

6.2 — Riemann Sums, Properties, and Integral MVT

1. Here $\Delta x = (2-0)/2 = 1$.

Rule	Sample inputs	Value
L_2	$0, 1$	$1(0^2+1^2) = \boxed{1}$
R_2	$1, 2$	$1(1^2+2^2) = \boxed{5}$
M_2	$1/2, 3/2$	$1[(1/2)^2+(3/2)^2] = \boxed{5/2}$

2. The widths are $2, 3, 1$, not all equal:

$$L = 2(1)+3(3)+1(4) = \boxed{15\text{ L}}, \qquad R = 2(3)+3(4)+1(5) = \boxed{23\text{ L}}.$$

Because the rate increases throughout the interval, the left sum is a lower estimate and the right sum is an upper estimate. The units are (L/min)(min) = L.

3. (a) Add adjacent intervals: $3+(-4) = \boxed{-1}$.

(b) Reverse the limits: $-\int_2^5 f = -(-4) = \boxed{4}$.

(c) Linearity and the length of $[0, 5]$ give

$$2(-1) - 6 + (5-0) = \boxed{-3}.$$

4. f is a polynomial, hence continuous on $[0, 3]$, so the integral MVT applies.

$$f_{\text{avg}} = \frac{12}{3-0} = \boxed{4}.$$

Solve $f(c) = 4$: $c^2+1 = 4$, so $c = \pm\sqrt{3}$. Only $\boxed{c = \sqrt{3}}$ is in $(0, 3)$.

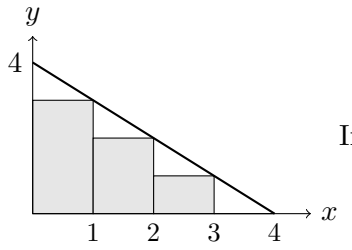
5. The triangle contributes $\frac{1}{2}(2)(3) = 3$; the semicircle has ordinary area $\pi/2$ but contributes negatively to the signed integral.

$$\text{(a) } \boxed{3 - \frac{\pi}{2}}, \qquad \text{(b) } \boxed{3 + \frac{\pi}{2}}, \qquad \text{(c) } \boxed{\frac{3-\pi/2}{4} = \frac{6-\pi}{8}}.$$

6. The integral is $0(1)+2(1) = 2$, so the average value is $\boxed{1}$. But the function takes only values 0 and 2, never 1. Thus **no such c exists**. There is no contradiction: f is discontinuous at $x = 1$, so it fails the continuity hypothesis of the theorem. Integrability alone does not guarantee attainment of the average.

6.3 — Inscribed/Circumscribed Rectangles and Limits

1. f is decreasing and nonnegative; width is 1. Right heights $3, 2, 1, 0$ give $\boxed{\ell_4 = 6}$. Left heights $4, 3, 2, 1$ give $\boxed{U_4 = 10}$.



Inscribed rectangles use right endpoints.

2. Now $\Delta x = 2/n$ and $f = x^2$ is increasing, so left gives lower and right gives upper:

$$\ell_n = \frac{2}{n}\sum_{i=1}^{n}\left(\frac{2(i-1)}{n}\right)^2 = \frac{8}{n^3}\frac{(n-1)n(2n-1)}{6} = \boxed{\frac{8}{3} - \frac{4}{n} + \frac{4}{3n^2}},$$

$$U_n = \frac{2}{n}\sum_{i=1}^{n}\left(\frac{2i}{n}\right)^2 = \frac{8}{n^3}\frac{n(n+1)(2n+1)}{6} = \boxed{\frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2}}.$$

3. Terms containing $1/n$ or $1/n^2$ tend to zero, so both limits are $\boxed{8/3}$. The area lies between these lower and upper sums for every n; the squeeze theorem forces its value to be $\boxed{8/3}$ square units.

4. (a) Width $3/n$, sample input $2 + 3i/n$, interval $[2, 5]$: $\boxed{\displaystyle\int_2^5 (1 + x^2)\,dx}$.

   (b) Width $\pi/(2n)$, sample input $\pi i/(2n)$, interval $[0, \pi/2]$: $\boxed{\displaystyle\int_0^{\pi/2} \cos x\,dx}$.

5. $\Delta x = \boxed{3/n}$ and $x_i = \boxed{1 + 3i/n}$, so

$$\boxed{\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{3 + \frac{3i}{n}}\frac{3}{n}}.$$

   The extra 2 inside $\sqrt{x+2}$ is added to the sample input $1 + 3i/n$.

6. On $[0, 1]$, the min/max are $0, 3$; on $[1, 3]$, they are $3, 4$; on $[3, 4]$, they are $0, 3$.

$$\ell = 1(0) + 2(3) + 1(0) = \boxed{6}, \qquad U = 1(3) + 2(4) + 1(3) = \boxed{14}.$$

   Both endpoint sums equal 9, not 14. On $[1, 3]$, the maximum 4 occurs at the interior point $x = 2$, whereas both endpoint heights are 3. Neither endpoint rule is an upper sum here.

6.4 — Fundamental Theorem of Calculus

1. (a) $[x^3 - x^2 + x]_0^2 = 8 - 4 + 2 = \boxed{6}$.

   (b) $[\ln x]_1^e = 1 - 0 = \boxed{1}$. These definite-integral values do not need $+C$.

2. The FTC yields $\boxed{A'(x) = \sqrt{1 + x^3}}$, hence $\boxed{A'(2) = 3}$. Differentiate once more using the chain rule:

$$\boxed{A''(x) = \frac{3x^2}{2\sqrt{1 + x^3}}}.$$

   The derivative of the integrand appears only in this *second* derivative.

3. (a) Substitute $t = x^3$ in the integrand and multiply by $(x^3)'$:

$$\boxed{G'(x) = 3x^2 \cos(x^6)}.$$

   (b) The variable is a lower bound, so $\boxed{H'(x) = -\ln(1 + x^2)}$.

4. Both bounds move:

$$\boxed{K'(x) = 3(1 + 9x^2) - 2x(1 + x^4)}.$$

   At $x = 1$, $K'(1) = 3(10) - 2(2) = \boxed{26}$. The lower-bound contribution must be subtracted, including its chain factor.

5. By the FTC, $F'(x) = x^2 - 4 = (x - 2)(x + 2)$ and $F''(x) = 2x$.

x	$(-3, -2)$	$(-2, 2)$	$(2, 3)$
$F'(x)$	$+$	$-$	$+$
F	increasing	decreasing	increasing

   Thus an interior local **maximum** occurs at $\boxed{x = -2}$ ($+$ to $-$), and an interior local **minimum** at $\boxed{x = 2}$ ($-$ to $+$). Since $F'' = 2x$, F is concave down on $\boxed{(-3, 0)}$ and concave up on $\boxed{(0, 3)}$. If values are desired, $F(x) = x^3/3 - 4x$, giving $F(-2) = 16/3$ and $F(2) = -16/3$.

6. Net mass change:

$$\int_0^4 (2t - 3)\, dt = [t^2 - 3t]_0^4 = \boxed{4 \text{ g}}.$$

   Therefore $\boxed{m(4) = 10 + 4 = 14 \text{ g}}$. The derivative $m' = 2t - 3$ is negative before $t = 3/2$ and positive after, so the mass is smallest at $\boxed{t = 3/2 \text{ min}}$.

6.6 — Numerical Integration: Approximations

1. The width is $h = 1$; endpoint values are $0, 1, 4$ and midpoint values are $1/4, 9/4$.

$$\begin{aligned} L_2 &= 1(0+1) = \boxed{1} && \text{low,} \\ R_2 &= 1(1+4) = \boxed{5} && \text{high,} \\ M_2 &= 1(1/4+9/4) = \boxed{5/2} && \text{low,} \\ T_2 &= \tfrac{1}{2}[0+2(1)+4] = \boxed{3} && \text{high,} \\ S_2 &= \tfrac{1}{3}[0+4(1)+4] = \boxed{8/3} && \text{exact.} \end{aligned}$$

   Comparison is with $I = 8/3$. The rectangle error directions follow from increasing behavior; midpoint/trapezoidal directions follow from concavity up. Simpson is exact here because the integrand is a quadratic.

2. The widths are $1, 2, 3$:

$$T = \frac{1}{2}(2+4) + \frac{2}{2}(4+3) + \frac{3}{2}(3+5) = 3 + 7 + 12 = \boxed{22 \text{ liters}}.$$

   Using a single common width would ignore the actual observation times.

3. Because $f' < 0$, f decreases:

$$\boxed{L_n \text{ over, } R_n \text{ under}}.$$

   Because $f'' > 0$, f is concave up:

$$\boxed{M_n \text{ under, } T_n \text{ over}}.$$

   A decreasing function can still be concave up; these are distinct properties.

A useful correction routine

For each approximation, check these in order:

1. Did I count intervals rather than data points?
2. Did I use the right widths and sample inputs?
3. Did I use the rule's correct weights and outside factor?
4. Does a justified under/over comparison agree with my arithmetic?
5. Did I distinguish the approximation from its error bound?

The remaining solutions continue on the next page.

6.6 — Numerical Integration: Error Bounds

4. For $f = x^4$ on $[0, 1]$,

$$f' = 4x^3, \quad f'' = 12x^2, \quad f^{(4)} = 24.$$

Thus the sharp derivative bounds are $\boxed{K_1 = 4,\ K_2 = 12,\ K_4 = 24}$. With $b - a = 1$ and $n = 4$:

$$E_L, E_R \le \frac{4(1)^2}{2(4)} = \boxed{\frac{1}{2}},$$
$$E_M \le \frac{12(1)^3}{24(4)^2} = \boxed{\frac{1}{32}},$$
$$E_T \le \frac{12(1)^3}{12(4)^2} = \boxed{\frac{1}{16}},$$
$$E_S \le \frac{24(1)^5}{180(4)^4} = \boxed{\frac{1}{1920}}.$$

These are guaranteed upper bounds, not claims that the actual errors equal them.

5. (a) Trapezoidal:

$$E_T \le \frac{3(2)^3}{12n^2} = \frac{2}{n^2} \le 0.001 \quad \Longrightarrow \quad n^2 \ge 2000.$$

Since $44^2 = 1936 < 2000$ and $45^2 = 2025 \ge 2000$, the smallest certified choice is $\boxed{n = 45}$.

(b) Simpson:

$$E_S \le \frac{10(2)^5}{180n^4} = \frac{16}{9n^4} \le 0.001 \quad \Longrightarrow \quad n^4 \ge \frac{16000}{9}.$$

The smallest integer satisfying the inequality is 7, but Simpson requires an even number of intervals. Since $6^4 = 1296 < 16000/9$ and $8^4 = 4096 > 16000/9$, choose $\boxed{n = 8}$. The phrase "smallest certified" refers to these bounds; the unknown actual error might be smaller.

6. (a) $I \in [2.4 - 0.03, 2.4 + 0.03] = \boxed{[2.37, 2.43]}$.

(b) Actual absolute error is $\boxed{|2.38 - 2.4| = 0.02}$. The percent error is

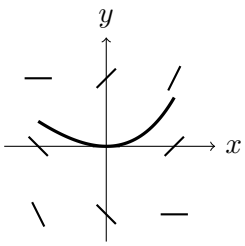
$$100\frac{0.02}{2.38} = \frac{100}{119}\% \approx \boxed{0.840\%}.$$

(c) A guarantee is an inequality, $|I - Q| \le 0.03$, not an equality. Here the actual error 0.02 is smaller than the allowed maximum.

7.1 — Slope Fields and Euler's Method

1. Substitute both coordinates into $x + y$:

	$x = -1$	$x = 0$	$x = 1$
$y = 1$	0	1	2
$y = 0$	-1	0	1
$y = -1$	-2	-1	0



One solution through $(0, 0)$ is $y = e^x - x - 1$ (formula not required).

The slope at $(0, 0)$ is $\boxed{0}$, but the solution is **not constant**: the candidate $y = 0$ would have derivative 0 while the equation requires $y' = x$ there. The sketched solution has a horizontal tangent and a local minimum at the origin.

2. $\boxed{y' = 2 - y}$ matches all three features. It is independent of x, so slopes repeat along rows; $2 - y$ has the required signs. The equilibrium is $\boxed{y = 2}$ since its slope is zero for every x.

3. With $h = 0.5$:

x_k	y_k	$x_k + y_k$	y_{k+1}
0	1	1	$1 + 0.5(1) = 1.5$
0.5	1.5	2	$1.5 + 0.5(2) = 2.5$

Thus $\boxed{y(1) \approx 2.5}$.

4. $h = (0 - 1)/2 = \boxed{-0.5}$.

x_k	y_k	$x_k - y_k$	y_{k+1}
1	2	-1	$2 + (-0.5)(-1) = 2.5$
0.5	2.5	-2	$2.5 + (-0.5)(-2) = 3.5$

Thus $\boxed{y(0) \approx 3.5}$.

5. (a) The updates are $0 + 0.5(2) = 1$, then $1 + 0.5(1) = 1.5$, then $1.5 + 0.5(0.5) = \boxed{1.75 \approx y(1.5)}$.

   (b) $y'' = -y' = -(2 - y) = \boxed{y - 2}$. This is negative while $y < 2$, so the solution is concave down there.

6. (a) One step: $1 + 1(1) = \boxed{2}$. (b) Two steps: $1 + 0.5(1) = 1.5$, then $1.5 + 0.5(1.5) = \boxed{2.25}$. (c) Both are below e. The absolute errors are approximately 0.71828 and 0.46828, respectively, so **two half-steps are closer**.

7.2 — Antiderivatives by Substitution

1. (a) $u = x^2 + 5$, $du = 2x\,dx$: $\int u^4\,du = u^5/5 + C = \boxed{(x^2+5)^5/5 + C}$.

   (b) $u = 3x + 1$, $du = 3\,dx$: $\int \cos u\,du = \boxed{\sin(3x+1) + C}$.

2. (a) $u = x^2 + 9$, $du = 2x\,dx$:

$$\frac{1}{2}\int \frac{1}{u}\,du = \boxed{\frac{1}{2}\ln(x^2+9) + C}.$$

   No absolute-value bars are needed because $x^2 + 9 > 0$.

   (b) $u = 1 + e^x$, $du = e^x\,dx$: $\boxed{\ln(1+e^x) + C}$; again the input to the logarithm is positive.

3. (a) $u = 2 + \cos x$, $du = -\sin x\,dx$:

$$-\int u^{-1/2}\,du = -2u^{1/2} + C = \boxed{-2\sqrt{2+\cos x} + C}.$$

   (b) $u = 3x^2$, $du = 6x\,dx$, and $9x^4 = u^2$:

$$\int \frac{1}{1+u^2}\,du = \boxed{\arctan(3x^2) + C}.$$

4. $u = x^2 + 5$, $du = 2x\,dx$; the limits $x = 0, 2$ become $u = 5, 9$:

$$\int_0^2 x\sqrt{x^2+5}\,dx = \frac{1}{2}\int_5^9 u^{1/2}\,du = \left[\frac{u^{3/2}}{3}\right]_5^9 = \boxed{\frac{27 - 5\sqrt{5}}{3}}.$$

5. Let $u = x^2$, $du = 2x\,dx$. Then $F = e^{x^2} + C$. From $F(0) = 3$, $1 + C = 3$, so $\boxed{F(x) = e^{x^2} + 2}$. Check: $F' = 2xe^{x^2}$ and $F(0) = 1 + 2 = 3$.

6. (a) Differentiating by the product or quotient rule gives

$$\frac{d}{dx}\left(\frac{\sin(x^2)}{2x}\right) = \cos(x^2) - \frac{\sin(x^2)}{2x^2}.$$

   The extra term means the claim is false.

   (b) $u = x^2$ gives $du = 2x\,dx$, but the integrand has no x factor to absorb. Dividing an antiderivative by a variable is not a valid reverse-chain-rule shortcut.

   (c) In the related integral, the needed factor *is* present:

$$\int x\cos(x^2)\,dx = \frac{1}{2}\int \cos u\,du = \boxed{\frac{1}{2}\sin(x^2) + C}.$$